

AS 9100 / IATF 16949 for PCBs

Aerospace + Automotive quality management — the hardest standards

What 'tier-1 supplier' actually means

Audience: Engineers working in automotive / aerospace supply chain

AS 9100 • IATF 16949 • PPAP • Traceability • Audits • Cost

Why These Standards

Safety-critical

Automotive + aerospace failures can kill.

Customer demand

Tier-1 OEMs (Bosch, Boeing, Tesla) require certification.

Liability protect

Audit trail defends in court.

Premium pricing

Certified suppliers charge more, justified by quality.

Process discipline

Certified processes mean less variability.

Global access

Same standards mean worldwide supplier qualification.

AS 9100 — Aerospace QMS

- ▶ Standard for aerospace + defense quality management
- ▶ Based on ISO 9001 + aerospace-specific additions
- ▶ Required by: Boeing, Airbus, Lockheed, BAE — every major aero OEM
- ▶ Certification body: SAE International + IAQG (International Aerospace Quality Group)
- ▶ Annual audits: stage 1 (documentation) + stage 2 (process)
- ▶ Cost: ₹3-15 L per year for cert maintenance (audit fees + internal QMS support)
- ▶ AS 9120: distributor variant; AS 9110: maintenance / repair variant
- ▶ Without AS 9100: cannot supply aerospace OEMs

IATF 16949 — Automotive QMS

- ▶ Standard for automotive industry quality management
- ▶ Based on ISO 9001 + automotive-specific additions
- ▶ Required by: every major car OEM (BMW, Daimler, Stellantis, Tata, Ola Electric)
- ▶ Certification body: IATF (International Automotive Task Force)
- ▶ Annual surveillance audits + 3-year recertification cycle
- ▶ Cost: ₹3-12 L per year
- ▶ Companion: AEC-Q components (passives + ICs qualified for automotive)
- ▶ Without IATF: cannot supply automotive tier-1 OEMs

PPAP — Production Part Approval

- ▶ PPAP = Production Part Approval Process
- ▶ Required by automotive OEMs before any part can ship at production volume
- ▶ 18-element documentation package: design records, drawing, materials, tests, etc.
- ▶ 5 levels of submission (Level 1: certificate only; Level 5: full part on hand)
- ▶ Level 3 is most common — full documentation + sample part
- ▶ Customer audits PPAP → approves OR rejects supplier
- ▶ Re-PPAP required for any design change or process change
- ▶ Cost: ₹2-10 L per part (engineering + sample + testing)

PPAP Elements (Sample)



Element	What it covers
1. Design records	Schematic, layout, BOM, drawings
2. Engineering change	ECN history, change approval
3. Customer engineering approval	Design sign-off
4. DFMEA	Design Failure Mode + Effects Analysis
5. PFMEA	Process Failure Mode + Effects Analysis
6. Control plan	What's checked at each step
7. Measurement system analysis	Repeatability of measurement equipment
8. Initial process studies	Production capability
9. Material requirements + testing	Substance test results
10. Performance testing	Sample units passing functional tests
11-15. Various reports	Sample, master, qualification, packaging
16-18. Submission docs	Package + warranty + customer approval

Traceability Requirements

- ▶ Every component lot traceable to date, batch, supplier
- ▶ Every solder paste lot traceable to date, batch
- ▶ Every PCB serial number traceable to fab + assembly batch
- ▶ Test results per serial number stored 7-15 years
- ▶ If field failure → identify which batch + retrace through QMS
- ▶ Recall: identify affected serial range from QMS database
- ▶ Documented via: ERP integration, MES, traceability software
- ▶ Cost of traceability infrastructure: ₹10-100 L for QMS + software systems

FMEA — Failure Mode + Effects Analysis

- ▶ DFMEA: design-level failure analysis
- ▶ For each potential failure: $\text{severity} \times \text{occurrence} \times \text{detection} = \text{RPN}$ (Risk Priority Number)
- ▶ $\text{RPN} > 100$ = action required to reduce risk
- ▶ PFMEA: same for production process
- ▶ Required for AS 9100 / IATF 16949 submissions
- ▶ Updated for every design / process change
- ▶ Output: corrective actions, design changes, process improvements
- ▶ Tool: dedicated FMEA software (APIS, IQ-Software) OR Excel templates

Audit Cycles

- ▶ Stage 1 audit (initial): documentation review
- ▶ Stage 2 audit (initial): on-site process review
- ▶ Annual surveillance: limited scope audit each year
- ▶ 3-year recertification: comprehensive re-audit
- ▶ Internal audit: required annually, must show closure of findings
- ▶ Customer audit: in addition to certifying body — customers audit their suppliers
- ▶ Non-conformance: must be closed within agreed timeframe (typically 30-90 days)
- ▶ Critical NC: certificate suspended until corrected

AEC-Q Component Qualification

- ▶ AEC-Q100: ICs qualified for automotive use
- ▶ AEC-Q200: passives (resistors, caps, inductors) qualified for automotive
- ▶ Test conditions: -40 to +125 °C operating; vibration, humidity, shock
- ▶ 1000-hour high-temp life test minimum
- ▶ Component vendor pays for qualification (₹50 L - 5 Cr per part)
- ▶ Automotive design REQUIRES AEC-Q parts on BOM
- ▶ Costs more (2-5×) than industrial-grade equivalents
- ▶ Without AEC-Q: part not acceptable for automotive applications

Cost of Compliance

- ▶ Initial certification cost: ₹5-30 L (audit + consulting + system setup)
- ▶ Annual maintenance: ₹3-15 L
- ▶ PPAP per part: ₹2-10 L (depending on complexity)
- ▶ AEC-Q components: 2-5× more expensive than industrial
- ▶ Traceability infrastructure: ₹10-100 L (ERP, MES, MES software)
- ▶ Internal QMS team: 1-5 engineers depending on volume
- ▶ Customer audit prep: ₹50k-2 L per major audit
- ▶ Total: small startup can't easily get certified; suppliers usually become certified as they scale

Path to Certification

- ▶ Step 1: Get ISO 9001 first (baseline QMS, ₹2-5 L cert)
- ▶ Step 2: Engage consultant familiar with AS 9100 / IATF 16949 (₹3-5 L)
- ▶ Step 3: Build QMS documentation (6-12 months of work)
- ▶ Step 4: Train all staff in new processes
- ▶ Step 5: Internal audits — catch issues before external audit
- ▶ Step 6: Stage 1 + Stage 2 audits by certifying body
- ▶ Step 7: If pass, receive cert (valid 3 years with annual surveillance)
- ▶ Total timeline: 12-24 months for first-time certification

Key Takeaways

Key takeaways from this lecture

Required for aero/auto

Without, no tier-1 OEM business.

PPAP per part

Major engineering effort.

Full traceability

Component → board → field, 7-15 years.

AEC-Q for auto components

Premium price.

12-24 month cert path

Plan ahead.

Annual surveillance

Continuous compliance.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ SAE International — AS 9100 standard
- ▶ IATF official site — IATF 16949 standard
- ▶ AEC-Q documents (AEC website)
- ▶ ISO 9001 — baseline QMS standard
- ▶ Local certification bodies: BSI, DNV, BV (Bureau Veritas)
- ▶ Indian QMS consultants — many offer end-to-end certification support
- ▶ PPAP templates from major OEMs (Bosch, Ford, GM)
- ▶ FMEA software: APIS, IQ-Software

Thank you • Build something this week.